

Future Temperature Changes in Java's Peri-Urban Landscapes

Ainur Ridho¹

Background / Context

Peri-urban areas, zones between urban and rural, in Java are highly vulnerable to rising temperatures due to rapid urbanization and climate change¹, but remain insufficiently studied. The difficulties separate peri-urban climates from urban or rural settings, hindering insights for planning and policy-making². Using Local Climate Zones (LCZs), this study maps historical and projected daily mean surface air temperatures (SAT) and extreme warm days (TG90p) to capture local variability in heat exposure.

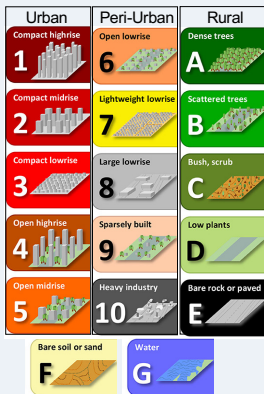
Objectives

1. How does temperature vary historically among different LCZs?
2. How will near-future (2030–2050) and distant-future (2080–2100) temperatures shift?
3. How will the frequency of extremely warm days (TG90p) change in peri-urban LCZs?

Methodology

Dataset	Variable	Original Resolution	Period	Purpose in Study
Local Climate Zones (LCZs) ³	Landform classification	~100 m	Static	Isolate landforms to analyze local temperature variability
Elevation (SRTM) ⁴	Topography (m)	30 m	Static	Support topography-based LCZ classification and contour mapping
Historical Observation (SA-OBS) ⁵	Surface Air Temperature (SAT) (°C)	~0.25° (~25 km), daily	1995–2014	Baseline temperature and calculation of historical TG90p
Climate Model (SINGV-RCM EC-Earth3) ⁶		~0.07° (~8 km), daily	2030–2050 & 2080–2100	Future temperature shifts under SSP1-2.6, SSP2-4.5, SSP5-8.5

Local Climate Zones⁷



*TG90p = # of days per year hotter than usual (top 10% of historical temperatures)⁸

Research Workflow

Data Preparation

- Regrid all datasets to ~8 km res.
- New LCZ classes x elevation: Lowland LCZ: <300 m, Highland LCZ: >300 m

Bias Correction

- Quantile mapping (QM) to historical model data to align with observation.
- Transfer QM correction to future projection models.

Main Analysis

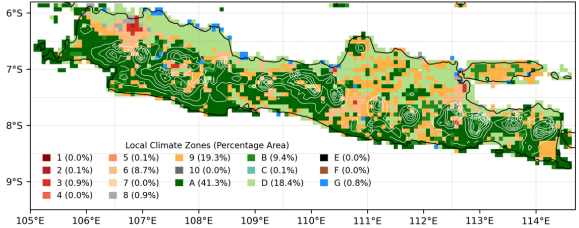
- Compute daily mean SAT distributions per lowland LCZ.
- Assign daily mean SAT (historical & projected) to its corresponding lowland LCZ.
- Identify extreme warm days (TG90p): days exceeding the historical 90th percentile for LCZ 6 lowland and 9 lowland (populated peri-urban).

Conclusions

- Daily mean SAT in Java is strongly shaped by land use and topography: urban and industrial LCZs retain the most heat, while peri-urban lowlands, though cooler historically, are increasingly vulnerable.
- Climate projections show significant warming across all LCZs, with peri-urban and lowland agricultural areas experiencing the largest increases (~2 °C by 2080–2100), consistent with global and regional urban heat studies.
- Extreme warm days (TG90p) in peri-urban LCZs are projected to rise from ~30 days/year in the near future to ~90 days/year in the distant future, highlighting peri-urban areas as emerging heat hotspots.
- These trends underscore the urgency for local and national adaptation strategies, including Indonesia's National Adaptation Plan (RAN-API), greening initiatives, heat-resilient infrastructure, and heat early warning systems targeting peri-urban communities.
- Integrating high-resolution LCZ-based climate data into planning can support evidence-based policies for disaster risk reduction, public health, and sustainable urban development, ensuring that peri-urban populations are not overlooked.

Findings

Local Climate Zones with Elevation Contours



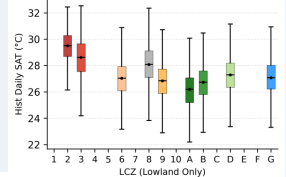
1

Urban zones retain more heat

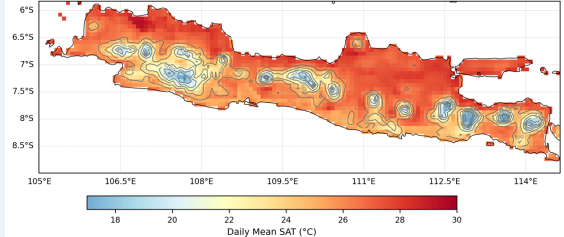
Daily mean SAT peak in urban LCZs (28–30°C) and industrial zones (28°C), while peri-urban and rural areas remain cooler (~27°C), emphasizing how dense built environments trap heat.

Lowland peri-urban zones are vulnerable
Most peri-urban LCZs (6 and 9) occur in lowlands, exposing dense settlements to heat due to limited elevation relief, similar to rural agricultural areas.

Historical (1995–2014)



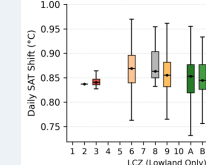
1995–2014 Historical Observation



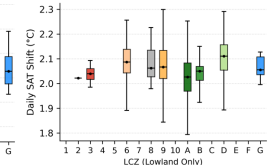
2

Near-future SAT rises across all LCZs, strongest shift in peri-urban and rural zones. By 2080–2100, warming reaches ~2 °C, largest in lowland agricultural and peri-urban areas, especially Tangerang, Bekasi–Cirebon corridor, Blora–Bojonegoro, and Parahyangan, Bromo–Semeru mountains.

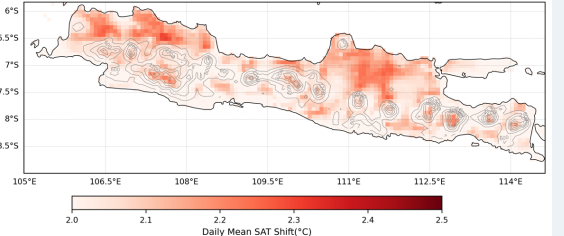
Near-Future (2030–2050) SSP2.45



Distant-Future (2080–2100) SSP2.45

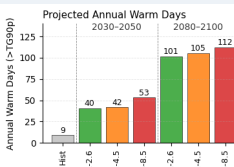


Distant Future (2080–2100) SSP2.45



3

In peri-urban areas (LCZ 6 and 9 lowland), the frequency of extreme warm days (TG90p) is projected to increase markedly, reaching roughly 30 days per year in the near future and up to 90 days per year in the distant future, with only minor differences across SSP scenarios.



References



¹World Resources Institute Indonesia